

SERVERLESS PORTFOLIO API USING AWS LAMBDA

Intern Details

- Name: SHANKER A
- Intern ID: CITS4540
- Domain: Cloud Computing
- Duration: 4 Weeks

1. Introduction

Serverless computing allows developers to build and run applications without managing servers. AWS Lambda is a serverless compute service that executes code in response to events and automatically manages the required infrastructure.

This project demonstrates the development of a serverless REST API using AWS Lambda and API Gateway.

2. Objective

To design and deploy a serverless API capable of providing student information, technical skills, project details, and contact information through multiple endpoints.

3. Technologies Used

- AWS Lambda
- Amazon API Gateway
- Python
- GitHub

4. Architecture

User → API Gateway → AWS Lambda → JSON Response

5. Implementation Steps

1. Created an AWS Lambda function.
2. Developed Python code for handling API requests.
3. Created API Gateway routes.
4. Connected API Gateway with Lambda.
5. Deployed the API.
6. Tested all endpoints.

7. Uploaded source code and documentation to GitHub.

6. API Endpoints

- GET /student
- GET /skills
- GET /projects
- GET /contact

7. Result

The serverless API was successfully deployed and tested. All endpoints returned JSON responses through API Gateway.

8. Conclusion

This project provided hands-on experience with AWS Lambda, API Gateway, serverless architecture, REST APIs, and cloud deployment. It demonstrated how cloud-native applications can be built without managing traditional servers.